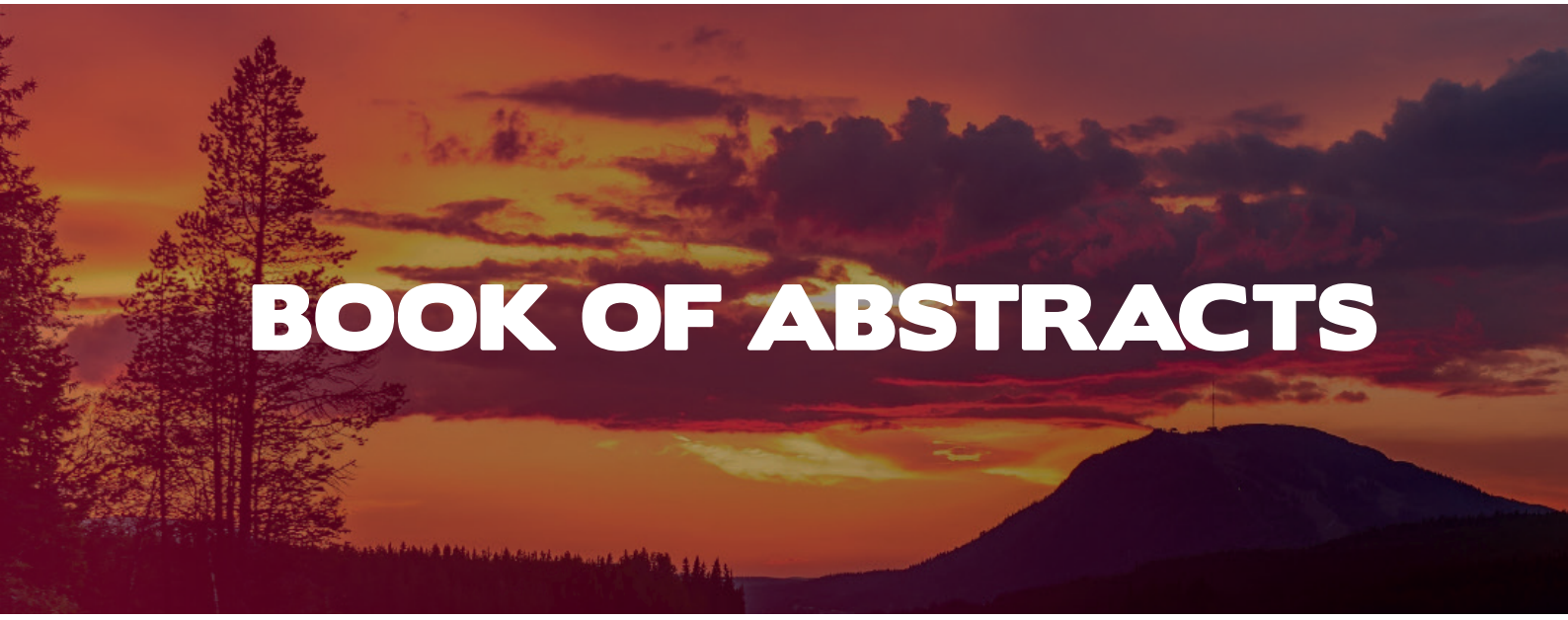
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Złoty Potok, May 8-10, 2026

INTERDISCIPLINARY DOCTORAL SYMPOSIUM



BOOK OF ABSTRACTS

RADA DOKTORANTÓW
Politechniki Wrocławskiej

Wrocław University
of Science and Technology

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Agricultural and Rural Development in Denizli, Turkey with Support from the IPARD Program

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Keywords: rural development, EU-IPARD, TOPSIS, fuzzy methods

This study assesses the EU Instrument for Pre-Accession Assistance for Rural Development (IPARD) on rural development in Denizli Province, Turkey. The aim of the study is to assess the level and spatial distribution of IPARD support in Denizli's rural districts and to examine its relationship with the socio-economic development profiles of these districts. The analysis integrates secondary data and primary survey data collected from IPARD beneficiaries (n =86) from a population of 723 projects, where the sample size was determined using a finite population correction method with a 10% margin of error, and interviews were allocated across districts using Probability Proportional to Size (PPS) sampling. The methodological framework combines multi-criteria decision-making and econometric approaches. A socio-economic development index was constructed using the TOPSIS method based on district-level indicators (H2), while agricultural structures were identified through fuzzy clustering techniques (H3). IPARD Satisfaction Index was developed using Triangular Fuzzy TOPSIS based on Likert-scale survey responses (H4). Hypotheses were tested using correlation analysis and ordinary least squares (OLS) regression (H1–H2), as well as ordered logit modelling (H4).

The results reveal a differentiated spatial distribution of IPARD support across districts. A statistically significant negative relationship is identified between socio-economic development and per capita IPARD support (H2), indicating that funds are more intensively directed towards less-developed districts, consistent with the program's objective. In addition, agricultural districts demonstrate higher engagement with the program in terms of support per hectare (H3), although investment intensity per project remains lower due to the prevalence of smaller-scale farms. At the individual level, the relationship between farm size and support is weak and statistically limited (H1). At the beneficiary level, the findings indicate that individual capabilities are key determinants of program success. Entrepreneurial initiative and participation in training activities significantly increase the probability of achieving higher IPARD success levels (H4), while education level is found to have a negative effect on the ordered logit model (H4). Cooperative membership is also negatively associated with performance outcomes (H4). Qualitative interview evidence supports this result, suggesting that many cooperatives function only formally ("signboard cooperatives") and remain largely inactive in practice. Furthermore, administrative burden and bureaucratic procedures are identified as key factors negatively affecting beneficiary satisfaction. Overall, the study demonstrates that IPARD operates as a territorially differentiated policy instrument, contributing both to rural development and to the redistribution of support towards structurally disadvantaged areas, while highlighting the importance of individual-level capacities and institutional effectiveness in maximizing program outcomes.

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Amyloid-Like Protein in the Enamel Matrix: Investigating Sequential and Structural Properties of Amelogenin

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Keywords: bioinformatics, amelogenin, amyloidogenesis, proteins structure modeling

Amelogenin is a protein found in the enamel matrix during the early stages of tooth development. The protein exhibits the ability to self-assemble into a stable scaffold for the developing enamel [1]. Microscopic studies show that the type of structure formed depends on the amino acid composition and the length of the amelogenin isoform. This is particularly important because the determinants of amyloid formation remain poorly understood, especially in terms of how sequence context, structural features, and environmental conditions influence different pathways of aggregation. Amelogenin, due to its ability to adopt different structural forms depending on isoform length and composition, provides a valuable model for identifying subtle yet critical factors that drive amyloid assembly. These structures may take the form of spherical nanodroplets or elongated, structured fibrils resembling amyloids [2, 3]. The formation process is fully controlled by protease activity, in contrast to pathological amyloidogenesis. Once their function is achieved, the development of the structure is discontinued, and it undergoes degradation [3]. To investigate the influence of specific protein motifs on the formation of different types of amelogenin aggregates, a sequence and structural analysis of 38 protein isoforms was performed. Results obtained using AGGRESCAN 4D, Waltz, and AlphaFold 3 enabled the identification of motifs, potentially involved in fibril formation. Interestingly, shorter protein isoforms, which according to the literature form nanodroplets, did not contain these fragments. The factors responsible for the formation of stable nanodroplets remain unknown, however, they are worth further investigation, particularly in the context of reports suggesting that similar structures may transition into amyloid fibrils [4]. The structural diversity and controlled nature of the aggregation process make amelogenin a model molecule for studying factors that determine differences in self-assembly behavior. At the same time, understanding the mechanisms of functional fibrillation may shed light on the processes distinguishing controlled from pathological amyloidogenesis, contributing to the development of research aimed at inhibiting disease-related processes.

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One- and Two-Photon Excited pH-Modulated Emission of Carbon Nanodots

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Keywords: carbon nanodots, multiphoton processes, fluorescence-based sensor, pH sensing

Carbon nanodots (CNDs) are a class of fluorescent nanomaterials of sizes smaller than 10 nm. They exhibit fascinating properties, including excellent photostability, biocompatibility, and tunability of light emission [1].

Non-toxic and biocompatible sensors are crucial for in vivo studies. In particular, fluorescent materials that can be excited via two-photon excitation have the upper hand over typical one-photon excited dyes due to precise localization and lower photon energy.

Carbon nanodots emitting cyan and yellow light have been developed by thermal decomposition of phloroglucinol, possessing high fluorescence quantum yields (exceeding 65% and 40%, respectively) and exhibiting two-photon excited fluorescence [2]. Here, we continue the studies on these nanodots, showing that their fluorescence intensity can be modulated by changes of the pH of the surrounding medium under both one- and two-photon excitation. This is the first report of such two-photon excited pH-modulated fluorescence over a wide pH range. These findings provide an excellent and sensitive material for applications in sensing, especially within the physiological pH range (6.5–8). Moreover, mechanistic studies, including UV-Vis spectroscopy, zeta potential and fluorescence lifetime measurements, were conducted in order to explain the mechanism of such pH dependence, aiding further understanding of the nature of this novel nanomaterial.

Acknowledgments

MC expresses gratitude to Dr. Marek Grzelczak and Dr. Ana Sánchez Iglesias. The authors acknowledge the funding from the Polish National Science Centre (OPUS 2024/55/B/ST5/02536).

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Age-Stratified EEG Synchronization in ADHD: Disentangling Disorder-Specific, Developmental, and Pharmacological Effects on Brain Network Dynamics

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Keywords: ADHD, EEG synchronization, functional connectivity, neurodevelopment, biomarkers

Attention-Deficit/Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental condition associated with widespread alterations in brain network organization. Electroencephalography (EEG) offers a non-invasive, temporally sensitive means of quantifying functional connectivity and large-scale network dynamics – features known to differ both in ADHD and across normal neurodevelopment. A critical, and frequently underestimated, methodological challenge is that many EEG synchronization signatures associated with ADHD overlap substantially with those arising from typical maturation. For instance, theta power decreases by up to 60% between early adolescence and adulthood in neurotypical individuals, meaning that a pattern indicative of disorder in one age group may fall within the normative range in another [1, 2]. Disentangling these trajectories is therefore a prerequisite for the development of new EEG based methods for ADHD detection.

My doctoral research addresses this challenge through a cross-cohort resting-state EEG paradigm spanning childhood through adulthood. Synchronization between EEG channels is quantified using windowed connectivity measures across the delta, theta, alpha, beta, and gamma bands, yielding dynamic synchronization matrices from which both node-level (mean connectivity, clustering coefficient) and network-level (global efficiency, spectral characteristics) graph-theoretic properties are extracted. Normative age-related trajectories of EEG synchronization are first established across matched healthy control groups spanning multiple developmental stages. These lifespan baselines then serve as reference functions against which ADHD-related deviations are evaluated – permitting the identification of synchronization changes that are genuinely disorder-specific rather than reflections of maturational lag or acceleration [3].

Building on this age-stratified framework, the study further isolates pharmacological effects by comparing connectivity profiles across cohorts that differ in stimulant medication status. Research indicates that commonly prescribed methylphenidate-class medications modulate theta/beta ratios and frontal coherence in ways that can partially normalize – or alternatively mask – disorder-related network signatures [4]. Regional interactions of particular interest include frontal-parietal and frontal–central connectivity, which have been implicated in attentional control and executive function deficits in ADHD across age groups [5].

The results are expected to yield age-stratified EEG synchronization norms, disorder-specific network biomarkers distinguishable from both healthy development and pharmacological modulation, and a framework for interpreting resting-state connectivity measures in the context of neurodevelopmental heterogeneity. These findings will contribute to the development of clinically applicable EEG-based tools for ADHD assessment across the lifespan.

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Spray Drying of Nano-YAG/SiC Composite Powders for Self-Healing Thermal Barrier Coatings

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Keywords: thermal spraying, atmospheric plasma spraying, TBC

Self-healing thermal barrier coatings (TBCs) are a promising candidate for extending the service life of gas turbine components operating at high temperature. Proper selection and preparation of feedstock powder materials are crucial for producing coatings with a controlled microstructure and effective self-repair mechanism [1].

This study focuses on the preparation of composite powders obtained by spray drying nano-sized yttrium aluminum garnet (YAG) with additions of submicron silicon carbide (SiC), intended as prospective feedstocks for plasma-sprayed thermal barrier coating (TBC) Top Coats. Conventional Top Coats based on yttria stabilized zirconia (YSZ) exhibit limitations in high-temperature stability, motivating the search for alternative materials. This work aims to develop and characterize YAG-based composite powder feedstocks with the potential for self-healing behavior, suitable for plasma spraying. Fine YAG powder with a median particle size of dv_{50} = 0.12 µm was used as the matrix material, while submicrometer-sized SiC powder (dv_{50} = 0.9 µm) was added at 6 wt.%; both pure YAG and YAG–SiC composite powders were prepared.

Self-healing in ceramic coatings can be achieved through oxidation-driven mechanisms, where high-temperature exposure leads to the formation of oxide phases that fill and seal microcracks. In this system, the self-healing capability is expected to arise from the oxidation of SiC, leading to the formation of silica-based phases that can seal microcracks in the coating.

The initial feedstocks were characterized using high-resolution scanning electron microscopy (HR-SEM) and laser granulometry to determine particle morphology and size distribution, followed by spray drying to obtain spherical, free-flowing powders. The resulting spray-dried powders will be characterized by HR-SEM to evaluate granule morphology. The outcomes of this work are expected to support further optimization of composite feedstocks for plasma spraying and their future application in advanced high-temperature coatings.

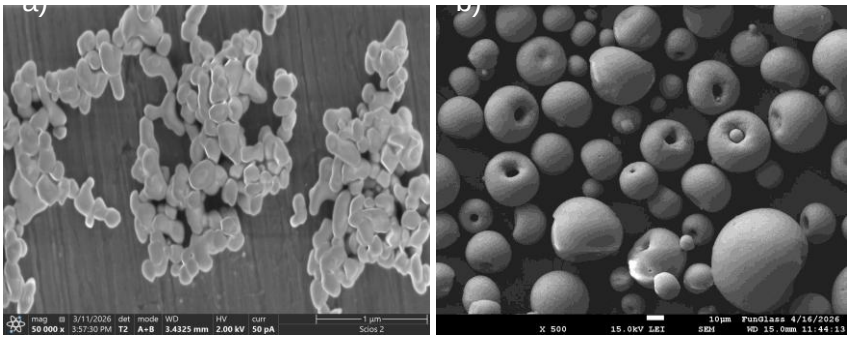


Fig. 1. SEM images of: a) initial YAG powder, b) spray-dried YAG-SiC powder

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Comparison of Rolling Bearing Fault Symptoms in PMSM and Induction Motors

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Keywords: rolling bearing faults, permanent magnet synchronous motor, induction motor, fault diagnostics, symptoms analysis

Electric motors are widely used in industry and electromobility, where they operate under diverse conditions and design configurations. They differ in construction, parameters, and operating regimes, which directly influences their behavior under fault conditions. Diagnostics of electrical machines is therefore essential to prevent costly downtime, severe failures, and potential safety hazards, while also supporting the development of more universal and transferable diagnostic systems.

Among various failure modes, rolling bearing faults are one of the most frequently occurring in electrical machines [1, 2]. However, the way these faults manifest differs between motor types. This work focuses on a comparative analysis of rolling bearing fault symptoms in permanent magnet synchronous motors (PMSM) and induction motors.

The study highlights that commonly used diagnostic signals, such as vibrations or stator currents, exhibit different fault-related features depending on the machine structure. In vibration spectrum analysis, the characteristic frequencies associated with bearing defects differ between motor types due to the presence of slip in induction motors, which is not present in synchronous machines such as PMSMs. Similarly, due to differences in motor construction, especially the presence of permanent magnets and the resulting magnetic field distribution, bearing faults generate different fault-related components and modulation patterns in electrical signals such as stator currents and stray flux.

The presented comparison aims to improve the understanding of fault symptom variability between motor types and to support the development of more universal diagnostic approaches.

Acknowledgments

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Photoinduced Electron Transfer in Stacked Nucleobases

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Keywords: photochemistry, electron transfer, nucleic acids, DNA, RNA, quantum chemistry, DFT, Marcus theory

Photoinduced electron transfer (PET) is a fundamental process underlying photosynthesis and photovoltaic technologies [1]. It can also occur between nucleotides in DNA and RNA strands, where it has been implicated in preventing and repairing damage induced by UV absorption [2]. Studying the influence of molecular features on PET in nucleic acids could lead to a better understanding of skin cancer development. It could also reveal features that will aid in the future design of more efficient organic solar cells.

We performed theoretical studies of PET in oligonucleotide sequences containing adenine (A) and guanine (G) nucleobases. The calculations were performed using the highly accurate ADC(2) method and time-dependent density functional theory (TD-DFT) with range-separated hybrid functionals, including CAM-B3LYP and ωB97X-D3. Electron transfer rates were estimated using Fermi's Golden Rule.

Our results revealed intriguing relationships between oligonucleotide composition and structure and the stability of the charge-transfer (CT) $\pi_G\pi_A^*$ states. We focused on structures with alternating A and G bases, such as AGA and GAG sequences. We demonstrated that in such structures, two CT $\pi_G\pi_A^*$ states can readily form, which were shown to be strongly coupled. This strong electronic coupling suggests that charge can be transferred very rapidly within these sequences, further stabilizing these states.

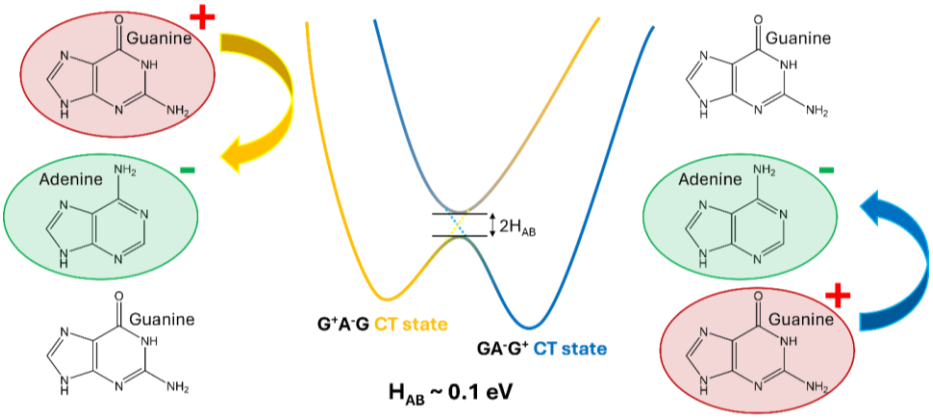


Fig. 1. Schematic representation of the two possible charge transfer states in the GAG structure and the two-state Marcus model for CT between these states. The calculated value of the H_{AB} electronic coupling between these states is also given

Acknowledgments

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In Vivo Quantification of Astrocyte-Dependent Metabolic and Neurotransmitter Changes Induced by Chronic Stress

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Keywords: astrocytes, glucocorticoids, metabolomics, major depression disorder, translational psychiatry

Astrocytes play a central role in regulating synapse number and function through their involvement in brain energy metabolism, neurotransmitter clearance, and neurotransmitter recycling. These processes are tightly interconnected and are regulated by systemic hormonal signals. Disruption of astrocyte-associated functions may therefore contribute to abnormal synaptic structure and signaling. Clinically observed features of depression include impaired glucose metabolism in the prefrontal cortex and altered glutamate/glutamine balance, both of which are closely linked to astrocyte activity. Astrocytes take up extracellular glutamate through specific transporters, thereby stimulating glucose metabolism; glutamate is then either converted to glutamine by the astrocyte-enriched enzyme glutamine synthetase or enters the Krebs cycle as an energy substrate. Current evidence suggests that these pathways are regulated by circadian metabolic signals, particularly glucocorticoids, under chronic stress conditions in mice [1]. However, it remains unclear how sustained glucocorticoid exposure reshapes extracellular metabolic dynamics across the day-night cycle.

The aim of this study is to determine whether chronic corticosterone administration alters astrocyte-associated extracellular metabolic and neurotransmitter-related profiles in a time-of-day-dependent manner. We hypothesize that chronic corticosterone exposure disrupts the rhythmicity of pathways related to glutamate-glutamine cycling, lactate metabolism, amino acid availability, and excitatory/inhibitory neurotransmitter balance. To test this, we will use in vivo NextGenMicrodialysis combined with targeted LC-MS/MS-based metabolomics to monitor temporal changes in approximately 120 selected polar metabolites in freely moving rodents. These will include lactate, branched-chain amino acids, amino acids, Krebs cycle intermediates, and neurotransmitter-related metabolites such as glutamate, GABA, serotonin, and glutamine. The analysis will focus on the prefrontal cortex, a key region involved in stress-response integration. Circadian effects will be evaluated by collecting microdialysate samples at defined Zeitgeber times across the light and dark phases, for example ZT3, ZT9, ZT15, and ZT21.

We expect to identify corticosterone-induced alterations in extracellular energy metabolism and neurotransmitter-related metabolic cycling, particularly within glutamate-glutamine-associated pathways. These changes may contribute to impaired neurotransmission in stress-related psychiatric disorders and reveal astrocyte-associated molecular targets for future therapeutic intervention in mood disorders.

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FFT-Based Load-Domain Completion for Broken Rotor Bar Diagnosis Under Unseen Operating Conditions

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Keywords: induction motor, broken rotor bars, motor current signature analysis, unseen operating conditions, load extrapolation

Reliable diagnosis of induction-motor faults becomes difficult when the operating conditions encountered during deployment are not represented in the training data [1]. This is particularly important for broken-rotor-bar faults, whose current signatures vary with load and may therefore change substantially across the operating range, while load-related effects are known to complicate rotor-fault interpretation in current analysis [2]. Motor current signature analysis for broken-rotor-bar diagnosis is already well established [3], and deep-learning-based diagnosis has also been increasingly explored in recent years [4]. However, explicit reconstruction of missing load regions for unseen-condition classification remains much less studied. This study investigates whether missing operating points can be reconstructed from measured current signals and then used to improve diagnosis under unseen-load conditions.

The proposed approach treats the problem as operating-domain completion. Three-phase stator current signals acquired for several rotor-bar damage levels and multiple load values are used to reconstruct missing loads in the Fourier domain, after which the generated signals are introduced into the training set. Performance is evaluated under a strict excluded-load protocol in which selected low-load or high-load regions are removed from the real training domain, and the final assessment is carried out only on real unseen-load data.

The results show that this strategy substantially improves classification under missing-load conditions. While the no-completion benchmark reaches a mean accuracy of 0.30 across the excluded-fold evaluations, FFT-based load completion increases this level to approximately 0.60-0.64, more than doubling the average performance. The strongest gains appear in the most demanding exclusion scenarios, indicating that the method is especially useful when the missing operating region is wide. The comparison of segment settings also shows that reconstruction quality depends on the adopted spectral granularity, so no single parameterization is universally optimal.

Overall, the study shows that load-conditioned spectral extrapolation can recover diagnostically useful information for operating conditions absent from the real training data. This suggests that operating-domain completion may provide a practical way to improve the robustness of current-based broken-rotor-bar diagnosis when load coverage in experimental datasets is incomplete.

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Hybrid ADRC-RBFNN Speed Controller Applied for a Drive with an Elastic Shaft

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Keywords: ADRC, Radial Basis Function Neural Network, Hybrid Speed Controller, Two-mass System, Electric Drive

Today, modern electric drives are facing strict industrial requirements. High dynamics combined with low maintenance costs and drive reliability are often demanded. In order to power various industrial devices, electric motors need to be coupled to external machinery. Due to the high complexity of different technological processes, the mechanical part of the drive often becomes more sophisticated. For the sake of scientific research, the combination of electric motor and external load machine can be substituted with two-mass drive system term. In fact, it represents the connection of above-mentioned devices, which are linked through an elastic component (shaft, clutch, belt, etc.). The described coupling is widely used in the field of robotics, industrial automation, automotive engineering, etc. However, because of the high complexity of the mechanical part of the drive, the issue of two-mass systems control constitutes a huge concern. Not only does the control system need to ensure satisfying plant dynamics, but it is also expected to provide torsional vibrations damping capabilities and to minimize the impact of torsional torque [1]. Classical methods deliver the expected response only when combined with multiple state space variable feedback. However, the above-mentioned approach features a strong dependency on plant parameters. In order to provide an eligible plant behavior under various operating conditions, adaptive control strategies are often used, though they require an additional reference model of the plant or the controlled process. To provide high resilience to different operating scenarios, robust control techniques are also often applied. To ensure the mentioned robustness and high dynamic properties an Active Disturbance Rejection Control (ADRC) is currently being tested in many applications. It ensures an eligible plant response (in a limited range) under different operating conditions, although the reference trajectory tracking may visibly decrease [2]. That is because the ADRC shares partial dependency on the plant parameters as well. The uncertain plant identification can lead to worse dynamic properties. Modern industry often relies on the implementation and deployment feasibility. Thus, each parametric deviation can be unfavorable. Therefore, to improve the viability of the ADRC algorithm, the research proposes a Hybrid-ADRC Speed Controller. The elaborated neural compensator has been tested for both nominal and significantly increased mechanical time constant of the load machine. It provides an eligible response even when plant parameters are significantly different than assumed. The algorithm mitigates speed overshoots and improves the system response when additional load torque occurs. Moreover, developed neural compensator does not require a sophisticated off-line training process, as the employed neural network uses on-line training technique. The proposed solution simplifies the deployment process of the drive, as the exact parameters of the flexible connection do not need to be precisely determined. They can be only roughly estimated. The plant uncertainty is compensated during drive operation. The general concept of the research involves elaboration of the current state of the art overview, development of a novel Hybrid-ADRC algorithm improvement based on neural network model, carrying out numerical and experimental tests. The proposed algorithm constitutes a huge contribution to the field of two-mass control strategies development.

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Control Methods of PMSM Electric Motors and Measurements of Dynamic States

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Keywords: PMSM, control, electric, motor, dynamic

This study aims to critically evaluate control methods for Permanent Magnet Synchronous Motors (PMSMs) used in electric and hybrid vehicle applications, with a particular focus on identifying the most effective strategies for achieving high efficiency and dynamic performance under real operating conditions. While the paper builds on existing literature, its main contribution lies in the integration of experimental observations with a comparative analysis of classical and modern control techniques, highlighting practical implementation challenges and performance trade-offs. The work concentrates primarily on vector control methods, especially Field-Oriented Control (FOC), and their practical realization using Clarke and Park transformations in conjunction with PI-based control structures. Experimental results, including measurements of back electromotive force, phase currents, and rotor speed under steady-state and transient conditions, are used to validate and contextualize the theoretical models. Additionally, the role of Space Vector Modulation (SVM) and cascaded feedback loops in improving torque and speed control accuracy is analyzed. To maintain focus, system-level aspects such as driving cycles and multi-component integration are discussed only insofar as they influence control performance and parameter variability. Based on the review and experimental analysis, FOC combined with SVM is identified as the most effective control strategy currently available, due to its superior decoupling of torque and flux, high dynamic response, and compatibility with digital control platforms. A sample motor control scheme is shown in Fig. 1.

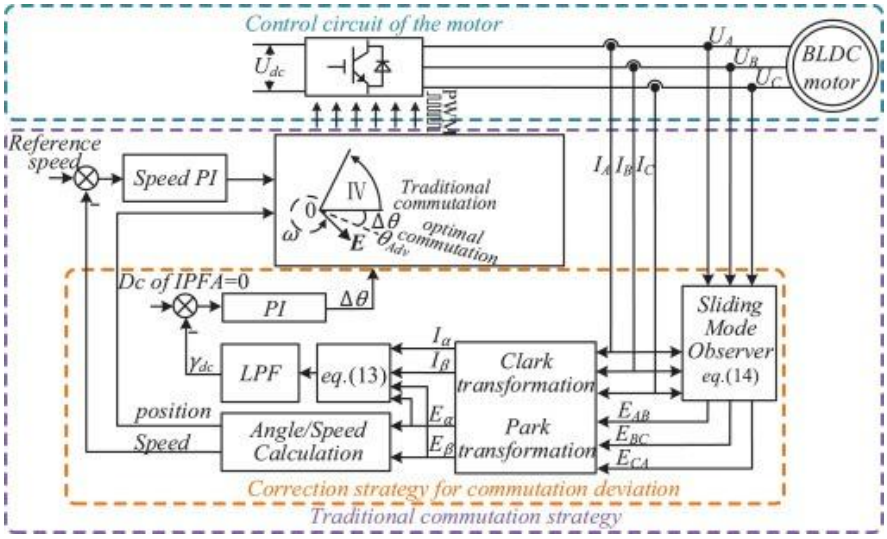


Fig. 1. Electric motor control Circuit Kun Liu, Wenjuan Zhang, Litao Dai, Jian Gao, Shoudao Huan

The study also identifies key limitations in current approaches, including insufficient integration of electro-thermal effects and limited automation in parameter identification. Future research directions are outlined, with emphasis on adaptive control methods and data-driven enhancements that can further improve robustness and efficiency in next-generation PMSM drive systems.

Sink-Float Separation as an Effective Method Supporting the Recycling of Polymer Materials in the Automotive Industry

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Keywords: flotation separation, plastics, automotive industry, circular economy

The circular economy aims to maintain materials in use for as long as possible. In the automotive sector, this necessitates the development of effective methods for the recovery and recycling of polymeric materials derived from end of life vehicles. Owing to the wide range of polymers used in automotive components, as well as the presence of modifying additives and changes in material properties resulting from service life, the production of homogeneous material fractions remains a significant technological challenge. One of the methods that may support this process is sink-float separation, which exploits differences in the density of polymer materials.

The aim of this paper is to evaluate the potential application of sink-float separation in the recycling of polymeric materials originating from the automotive industry and to determine its relevance to the implementation of circular economy principles. In this study, sink-float separation was experimentally verified using real polymer waste obtained from end of life automotive headlamps, followed by the separation of selected density fractions and determination of their mechanical properties. The paper outlines the fundamentals of the sink-float process, with particular emphasis on the role of density differences, medium selection and the choice of process parameters enabling the efficient separation of polymer fractions. The analysis was experimental and included shredding of the waste, particle size evaluation, density-based separation in potassium hydroxide solutions, injection moulding of test samples, and tensile testing. Two separated fractions were analysed: C1, with a density above 1.172 g/cm^3, and C2, with a density of 1.113 to 1.172 g/cm^3.

The analysis indicates that sink-float separation can effectively support the recovery of polymeric materials from the automotive sector by increasing the purity of the recovered fractions and enhancing their suitability for further processing. The obtained quantitative results showed differences in the mechanical behaviour of the separated fractions. Fraction C2 achieved higher stiffness and tensile strength, whereas fraction C1 showed slightly higher elongation at break. From a circular economy perspective, this technology contributes to reducing the consumption of virgin raw materials, decreasing the volume of waste, and extending the service life of materials used in the automotive industry. However, its effectiveness depends on appropriate feed preparation and the optimisation of process conditions. The main technological limitation of this method is the difficulty in separating polymer materials with very similar density values. As a result, sink-float separation may be ineffective for such materials, and additional separation processes, such as electrostatic separation, may be required to improve the purity of the recovered fractions.

Gasochromic Response of Pt/WO_3 Thin Films at Low Hydrogen Concentrations

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Keywords: gasochromic, tungsten oxide, thin films

Hydrogen is a key component of a sustainable energy future. At room temperature, it exists as a colorless, odorless, and highly flammable gas (H_2). Its wide flammability range, extremely low ignition energy, and nearly invisible flame pose serious safety risks, including the potential for sudden explosions [1]. Among the investigated methods of hydrogen detection, gasochromic sensors offer safer operation, greater flexibility and exceptional selectivity, as the gasochromic effect of tungsten oxide is specific to hydrogen [2]. The aim of this study is to show that tungsten oxide films deposited by magnetron sputtering provide sufficient sensitivity and stability for the detection of low concentrations of hydrogen.

Tungsten trioxide thin films were fabricated using magnetron sputtering. Gasochromic properties of WO_3 thin films was measured in the presence of thin platinum catalyst (Pt/WO_3). To measure the change in optical transmission H_2/Ar mixture was supplied for 30 minutes, followed by air for another 30 minutes to restore the sample's transparency. Based on the transmission change measurements (Fig. 1a) for the specified wavelengths, a variation in transmittance was detected for wavelengths exceeding 600 nm, with the most substantial change occurring at approximately 900 nm. For a wavelength of 900 nm, variations in transmittance were determined for hydrogen concentrations in the range of 25–100 ppm (Fig. 1b). During the colouring process, it was observed that the relative change in transparency ranged from around 84% for 25 ppm to around 92% for 1000 ppm.

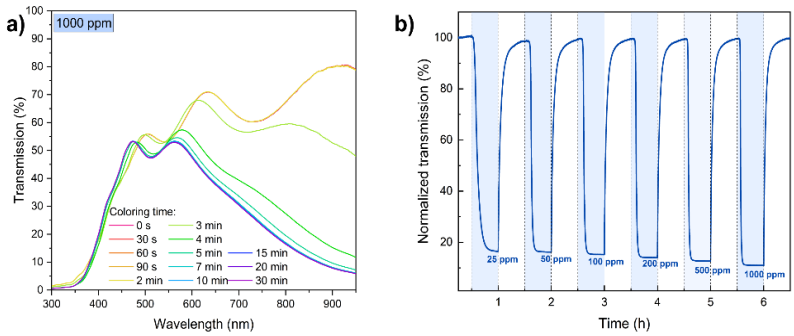


Fig. 1. a) Transmission spectra, b) normalized transmittance values of WO_3 film during exposure to hydrogen

These findings confirm that the Pt/WO3 thin films fabricated by magnetron sputtering provide a rapid and significant gasochromic response, confirming their capability for sensitive and selective detection of low hydrogen concentrations.

Acknowledgments

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Boosting EP-SCGD OES Signals: The Role of Surfactant and Pressure

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Keywords: elevated pressure, multielement analysis**,** non-ionic surfactant**,** solution cathode glow discharge

Modern analytical chemistry strives to miniaturize and reduce the cost of measurement systems while maintaining the sensitivity offered by advanced techniques such as ICP OES and ICP MS. One of the most promising directions is solution cathode glow discharge optical emission spectrometry (SCGD OES), first introduced by Cserfalvi and Mezei in 1993 [1]. Although this method is valued for its simplicity and compactness, its analytical performance is sometimes limited by inefficient analyte transport from the solution to the plasma. The method is particularly suitable for trace metal determination in environmental and biological liquid samples.

Since elevated pressure is known to significantly influence plasma characteristics and discharge stability [2], this study systematically investigates a strategy combining elevated gas pressure with chemical modification of the liquid cathode to improve metal detection performance. Particular attention was given to the synergistic effect of a nonionic surfactant, Triton X-405, and elevated pressure (up to 1.6 atm). The mechanism is based on modifying the physicochemical properties of the solution, namely decreasing dynamic surface tension, as previously reported for surfactant-assisted SCGD systems [3]. These changes reduce water evaporation and stabilize the plasma discharge, leading to more efficient transport of metal species into the excitation region.

Under optimized conditions (surfactant concentration of 4.1 mmol L^{-1} and helium pressure of 1.6 atm), a significant enhancement of atomic emission signals was observed. Detection limits (DLs) for the EP-SCGD OES system ranged from 0.02 to 6 µg L^{-1}. These results are comparable to conventional ICP OES instrumentation and, in several cases, superior. Notably, the DL for Rb (0.02 µg L^{-1}) was more than three orders of magnitude lower than that obtained with ICP OES (70 µg L^{-1}). Significant improvements were also noted for elements such as Ag, Hg, and Pb, demonstrating the synergistic effect of elevated pressure and surfactant addition.

The developed protocol significantly improves the analytical capabilities of SCGD OES, making it a cost-effective alternative to more complex instrumentation. Overall, the study demonstrates that simultaneous control of plasma conditions and solution chemistry enables measurable gains in sensitivity and detection limits, indicating that compact SCGD systems can approach the performance of established ICP-based techniques.

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Homometallic Zinc Molecular Precursors for Synthesis of Heterometallic Rare Earth Complexes

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Keywords: rare-earth elements, lanthanides, metal complexes, oxide matrices, high-level waste

Multimetallic synergy and cooperativity in heterometallic rare earth complexes have attracted attention over the past three decades due to their applications as enantioselective catalysts, sensors, optical devices, and in bio-imaging. Typically, these compounds are synthesized by dissolving simple metal reagents in water–alcohol mixtures with amines used to deprotonate phenolic ligands. This approach yields thermodynamically stable crystalline products whose composition depends on reaction time, conditions, and reagents. However, the structural and compositional diversity of the obtained complexes complicates further studies, so most research focuses on characterizing only three or four isostructural complexes.

Unlike most of the published papers, our approach was to develop universal methods for obtaining heterometallic rare earth – zinc complexes using homometallic zinc precursors. We developed a simple synthetic route to RE^{III}–Zn^{II} coordination compounds by reacting $[Zn_4(L)_8]$ and $[Zn_4(\mu_3\text{-}OMe)_2(L)_6]$ (HL = methyl salicylate) with 2 equiv of $RECl_3$. This approach led to the formation of trinuclear $[REZn_2(L)_4Cl_3(H_2O)]$ (RE^{III} = Y, Ho, Er) or hexanuclear $[RE_2Zn_4(\mu_3\text{-}OH)_2(\mu\text{-}OMe)_2(L)_6Cl_4]$ (RE^{III}= Y, Ho, Er) heterometallic aryloxido clusters, as pictured in Fig. 1.

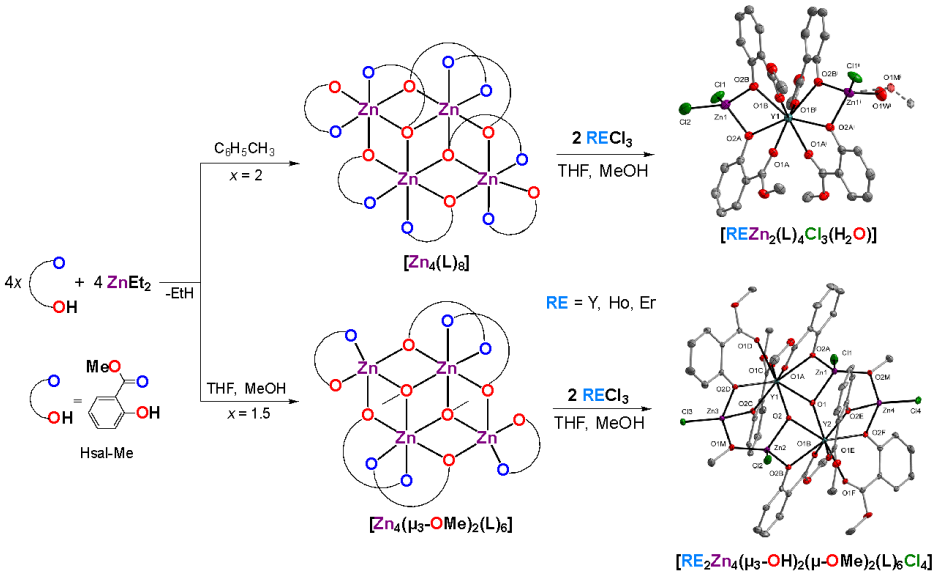


Fig. 1. Synthesis of $[REZn_2(sal\text{-}Me)_4Cl_3(H_2O)]$ (RE^{III} = Y, Ho, Er) and $[RE_2Zn_4(\mu_3\text{-}OH)_2(\mu\text{-}OMe)_2(sal\text{-}Me)_6Cl_4]$ (RE^{III} = Y, Ho, Er) heterometallic aryloxido clusters

Acknowledgments

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Effect of Temperature and Composition on the Electrical Properties of FETs Based on $Mo(S_XSe_{1-X})_2$ Alloys

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Keywords: 2D materials, FET, transition metal dichalcogenides, MoSSe alloys

In the field of 2D materials, significant progress has been made in semiconductor research. A prominent example is transition metal dichalcogenides (TMDs), which exhibit high electron mobility and stability, a thickness-dependent bandgap, high on/off current ratios (I_{ON}/I_{OFF}), and reasonable hole mobility. They demonstrate excellent electrostatic control and improved resistance to short-channel effects beyond the capabilities of silicon-based technologies [1]. Furthermore, the fabrication of TMD alloys enables tuning their optical, electrical, and mechanical properties, for example, to enhance field-effect transistors (FET) performance [2]. To fabricate the FET, specially prepared SiO_2/Si substrates with planar gold electrical contacts were used. The results of the research indicate that modifying the S/Se ratio in the $Mo(S_XSe_{1-X})_2$ alloy improves FET efficiency, as shown in the figure below. For instance, the I_{ON}/I_{OFF} ratio, for $MoS_{0.82}Se_{1.18}$ increases by one order of magnitude compared to the parent TMDs (i.e., MoS_2 and $MoSe_2$). For digital logic applications, the I_{ON}/I_{OFF} ratio should exceed 10^5 [3]. The research shows that the use of alloys makes it possible to meet this requirement at temperatures (T) in the range of 350–400 K. The similar trend can be observed for transconductance (g_m), which is highest at T = 350 K and lowest at T = 80 K and T = 450 K. Furthermore, for the $MoS_{0.82}Se_{1.18}$ alloy at room temperature, g_m is one order of magnitude higher than that of MoS_2 and $MoSe_2$. It should also be noted that the electrical properties of the FET change nonlinearly with increasing temperature, as shown in the figure below. Based on these preliminary results, it can be concluded that the electrical properties of FETs are strongly temperature-dependent and that modifying the S/Se ratio in the $Mo(S_XSe_{1-X})_2$ alloy affects both device performance and thermal stability.

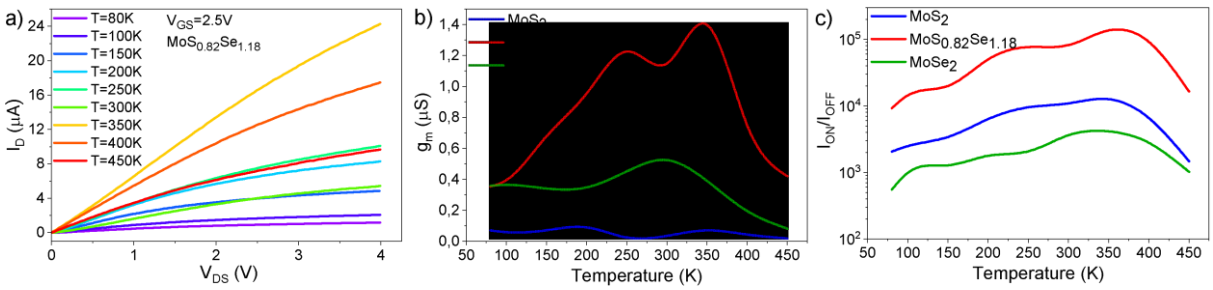


Fig. 1. Temperature dependence of the FET properties: (a) output characteristics of $MoS_{0.82}Se_{1.18}$ FET measured at a constant gate-source voltage (V_{GS} = 2.5 V), (b) the transconductance (g_m), and (c) the ratio of on-state to off-state current (I_{ON}/I_{OFF})

Acknowledgments

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Design and Properties of New Multilayered Perovskites

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Keywords: perovskite, 2D structures, Ruddlesden–Popper, Optoelectronics, Non Linear Optics

Multilayered Ruddlesden–Popper (RP) perovskites represent a promising platform for the development of multi-functional optoelectronic materials. Their unique layered architecture enables independent tuning of structural and electronic properties through the choice of organic cations and the control of dimensionality [1, 2].

Here, we discuss our goals and approach to obtain a family of hybrid lead halide perovskites incorporating selected organic cations that can act both as a spacer and perovskite cage cation. By systematically changing the type and ratio of organic cations it is expected to obtain different RP phases (*n*) [3, 4]. By using multi-technique approach incorporating optical and dielectric spectroscopy, X-ray diffraction and optical analysis this work seeks to uncover structure–type relationships induced by mixed cations, and to clarify the structure-directing role of individual cations.

Our preliminary results suggest that these systems can exhibit appealing optical properties, including nonlinear optical behavior, depending on the crystal structure. In the future we aim to obtain and characterize new 2D perovskites and pinpoint how different cation additives can induce desired crystal phases.

Such work is expected to highlight the role of choosing a cation and cations ratio as a key strategy for designing layered perovskites and expanding the knowledge about engineering such structures.

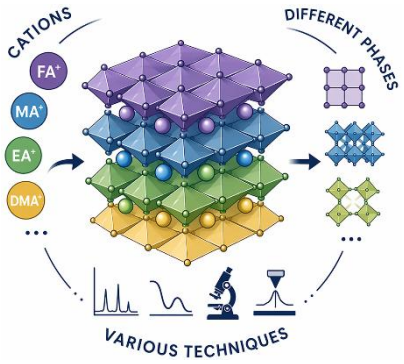


Fig. 1. Analyzing cation-structure relationships

Acknowledgments

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Computational Investigation of Binding Patterns of Non-Standard RNA Base Pairs

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Keywords: RNA, nonenzymatic, self-replication, 2-thiouridine, binding energies

Accurate replication of nucleic acids is essential for preserving genetic information, as incorrect incorporation of a nucleotide may lead to mutations and even cancer [1, 2]. Eukaryotes have protein based systems in place for recognition and repair of incorrectly incorporated nucleotides or resynthesize the DNA strand. However, in the prebiotic era, such advanced protein machinery was absent, and other mechanisms must have been responsible for reducing mutation rates in the RNA self-replication [3, 4].

Nonenzymatic RNA primer extension has a ~17% misincorporation rate, largely due to the guanosine–uracil wobble pair.[5] Non-canonical nucleotides, such as inosine or 2-thiouridine, may form more stable and accurate base pairs [6, 7]. Substitution of uracil with 2-thiouracil increases replication efficiency without significantly altering the incorporation of subsequent nucleotides. Interestingly, 2-thiouracil can form a strong homobase pair (tU:tU), which stabilizes the double-stranded RNA helix [8]. However, the tU:tU base pair exhibits a strong stalling effect, preventing further primer extension after incorporation of tU opposite tU.

Molecular dynamics simulations and DFT calculations allow investigation of the influence of 2-thiouracil on nonenzymatic RNA self-replication at the molecular level. Clustering algorithms reveal behavioral patterns indicating how effectively activated mononucleotides remain bound to the binding site, while DFT binding energy calculations show that reaction rates cannot be assessed solely based on the binding pattern of a single base pair. Instead, the conformational stability of the activated mononucleotide and the corresponding transition state is a key factor governing the efficiency of the process, rather than binding energy alone. In conclusion, evaluation of nonenzymatic RNA primer extension requires consideration of the overall double-helix structure and the conformational stability of the activated mononucleotide.

Acknowledgments

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Using Unlearning Based Approach for Continual Learning Concept Drift

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Keywords: continual learning, concept drift, machine unlearning, data streams, machine learning

Traditional machine learning [1] assumes a stationary data distribution, yet many real-world applications operate on nonstationary streams in which the underlying concept evolves over time. This problem can also be viewed as task-free continual learning under concept drift, where a model must adapt sequentially without explicit task identities or task boundaries. In such settings, effective learning requires both rapid adaptation to new data and forgetting of outdated information. A common solution is based on a sliding window [2], but this approach is often computationally demanding because the model must be repeatedly retrained from scratch on the most recent data. In this seminar we will present a different perspective based on machine unlearning [3] called UIL (**U**nlearned and **I**teratively trained c**L**assifier). Instead of rebuilding the model each time the active window changes, we remove the influence of outdated samples using unlearning and then update the model with newly observed data. This enables efficient, targeted forgetting while preserving adaptation to evolving distributions. As a base unlearning algorithm, we used Certified Data Removal [4]. Empirical results on image stream classification across multiple drift scenarios demonstrate that the proposed approach offers a competitive and computationally efficient alternative to standard sliding-window retraining. Using recovery analysis [5], we detected significant improvement in maximal and average deterioration values. UIL generally reduces the worst deterioration relative to SW. UIL consistently uses an average of 8 MB less per batch than SW, although this difference is modest compared to the reduction in runtime, from 6.28 s per batch for SW to 2.2 s for UIL. It is also worth noting that, for most tested parameter values, UIL remains stable, with instability observed only at very high learning rates. This result is a huge milestone, and to the best of our knowledge, this is the first successful attempt to connect machine unlearning with concept drift mitigation for task-free continual learning. Future studies may include going deeper into this area towards controlling unlearning with drift detectors and using different unlearning algorithms in order to find the optimal one for this task.

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Do You Trust Me?
The Hidden Math Behind Sharing Personal Data with Brands

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Keywords: Hyper-personalization, Privacy Calculus, Artificial Intelligence, Consumer Trust, Marketing

The research aim focuses on the mechanisms through which marketing hyper-personalization and brand trust influence consumer willingness to share data within the Privacy Calculus paradigm. This theory posits that individual decisions result from a rational calculation between perceived benefits and the risks associated with privacy violations. While existing literature has primarily addressed general forms of marketing communication and global attitudes toward privacy, the rapid advancement of Artificial Intelligence and Big Data analytics is fundamentally redefining these relationships.

Modern hyper-personalization, characterized by unprecedented scale and precision in processing behavioral data, introduces a new dynamic into the consumer decision-making process. Consequently, a significant research gap emerges regarding the extent to which the tangible benefits of highly individualized messaging can offset privacy concerns and determine the propensity to disclose sensitive information. These premises justify the need for an empirical verification of the decision-making mechanisms of Polish consumers in the face of hyper-personalization, situated within specific cultural and legal frameworks.

Legal Personality for the Odra River

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Keywords: legal personality of nature, rights of rivers, ecocentrism vs anthropocentrism, non-human legal subjectivity, comparative environmental law

The Bill on the Recognition of the Legal Personality of the Oder River (Parliamentary Print No. 2082, 10th term of the Sejm of the Republic of Poland) [1] constitutes a pioneering legislative proposal aimed at conferring upon one of Poland's largest river ecosystems the status of an independent legal subject. The immediate genesis of this regulation is the ecological catastrophe of the summer of 2022, in which – according to various estimates – more than 215 million aquatic organisms perished. The 2023 report of the Supreme Audit Office (NIK) [2] revealed egregious deficiencies in water management in Poland, demonstrating that the existing mechanisms of legal protection were structurally inadequate. The bill originates from the citizens' legislative initiative of the "Clean Oder" Committee (Komitet "Czysta Odra"), which was not proceeded as a citizens' initiative due to the failure to collect the requisite 100,000 signatures within the statutory time limit – only 92,998 signatures were deemed valid. The substantive content of the original initiative has been preserved in its entirety in the subsequent parliamentary bill introduced by members of the Sejm. The fundamental legal premise of the bill is the reclassification of the river from the category of a legal object to that of a legal subject, endowed with a catalogue of fundamental rights – including the right to exist, to flow freely, to regenerate its resources, to be protected from pollution, and to obtain compensation for damages sustained – and represented by a fifteen-member Oder Representative Committee (Komitet Reprezentantów Odry). As the authors of the bill themselves point out, "[t]he purpose of the bill is to ensure effective legal protection for the entire ecosystem of the Oder River and to establish mechanisms for inter-institutional cooperation to ensure the broadly understood health and well-being of the river. [...] With the introduction of a clear catalog of fundamental rights, the assurance of broad and diverse representation (Committee of Representatives), the incorporation of the most up-to-date research findings regarding the Oder (Scientific Committee), as well as the restoration of the Ministry of the Environment's rightful role, there will be a real chance to change the current situation, which threatens the safety of people and the ecosystem". This solution draws upon analogous regulatory frameworks adopted by New Zealand (Whanganui River, 2017), Colombia (Atrato River, 2016), and Spain (Mar Menor lagoon, 2022), and is consistent with the Water Framework Directive (2000/60/EC) as well as EU Regulation No. 2024/1991 on nature restoration. The innovative character of the bill lies in its replacement of the existing administrative-repressive model with a preventive mechanism grounded in the principle of proportionality and structured around negotiated agreements with the Representative Committee, supported by an independent Scientific Committee (Komitet Naukowy Odry). In this manner, the bill introduces into the Polish legal order the paradigm of "rights of nature" as an instrument of effective environmental protection. On March 27, 2026, during the first reading, the bill was referred to the Committee on Maritime Affairs and Inland Navigation and the Committee on Environmental Protection, Natural Resources, and Forestry.

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When Bacteria Meet Amyloid – the Role of Hpn Protein in Aβ Aggregation

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Keywords: humane microbiome, Helicobacter Pylori, Alzheimer's disease, histidine-rich protein

Amyloid formation is a hallmark of neurodegenerative disorders, including Alzheimer's disease (AD), which is characterized by the aggregation of β-amyloid (Aβ) peptides and tau protein. Amyloids are ordered fibrillar aggregates with a cross-β structure, formed through protein misfolding processes that remain poorly understood [1]. Although typically associated with human pathology, amyloid structures can also be produced by microorganisms. One such example is *Helicobacter pylori*, a well-known gastric pathogen implicated in ulcer disease. Increasing evidence suggests a link between *H. pylori* infection and the incidence of AD [3]. Notably, the presence of this bacterium has been associated with altered levels of Aβ in both cerebrospinal fluid and peripheral blood [4].

H. pylori secretes a Hpn protein, which has been shown to assemble into amyloid-like fibrils in its recombinant form [2]. Hpn is particularly unusual due to its high hydrophilicity and exceptionally high histidine content (47%), distinguishing it from typical aggregation-prone proteins. Under certain pathological conditions, the integrity of the blood-brain barrier may be compromised, and in vitro studies indicate that Hpn is capable of crossing this barrier [3], [5]. Additionally, the gut-brain axis may provide a route for interactions between Hpn and Aβ [3]. Taken together, these observations suggest a potential mechanistic link between *H. pylori* infection and AD. Specifically, interactions between Hpn and Aβ may promote amyloid aggregation and enhance plaque deposition in the brain, thereby contributing to disease progression.

The aim of this study is to evaluate the aggregation propensity of Hpn and its potential interactions with Aβ using a combination of experimental techniques, including atomic force microscopy (AFM), electron microscopy (EM), thioflavin T (ThT) fluorescence assays, Fourier-transform infrared (FTIR), and circular dichroism (CD) spectroscopies [1], [6]. Demonstrating a direct interplay between Hpn and Aβ may provide new insights into the mechanisms underlying the potential contribution of *H. pylori* to AD pathology and could aid in the identification of novel therapeutic targets.

Acknowledgments

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Impact of Silver and Gold Nanoparticles on Methylene Blue Photostability and ROS Generation for Photodynamic Therapy

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Keywords: photodynamic therapy, methylene blue, silver nanoparticles, gold nanoparticles, singlet oxygen generation, photobleaching

The clinical efficacy of photodynamic therapy (PDT) is fundamentally determined by the efficient generation of reactive oxygen species (ROS) upon light activation of a photosensitizer (PS). Methylene blue (MB) is a widely utilized phenothiazinium dye in PDT; however, its practical application is often limited by poor photostability and low quantum yields of singlet oxygen ($^1O_2$) in complex biological environments. Plasmonic nanoparticles exhibit Localized Surface Plasmon Resonance (LSPR) – upon light irradiation, conduction electrons oscillate collectively, generating intense local electromagnetic fields. When LSPR overlaps with MB absorption, near-field enhancement may amplify ROS generation by modulating PS excited triplet state. The objective of this study was to investigate the influence of silver nanoparticles (AgNP) and gold mini-nanorods (miniAuNR), on the photochemical performance and stability of MB.

The experimental procedures involved the preparation of MB-nanoparticle hybrid systems and their subsequent characterization under controlled irradiation conditions. Photobleaching kinetics were monitored using UV-Vis spectrophotometry to evaluate the degradation rate of the dye in the presence of AgNPs and miniAuNRs. Furthermore, the efficiency of singlet oxygen production was quantified using Singlet Oxygen Sensor Green (SOSG) as a fluorescence probe, allowing for a comparative analysis of ROS generation dynamics.

Our results demonstrated that the integration of nanoparticles significantly altered the photochemical properties of MB, with miniAuNRs exhibiting superior performance compared to silver-modified systems. We also found that the addition of miniAuNRs reduced the photobleaching rate of the dye by 90%, significantly enhancing its operational stability. Finally, a 200-fold increase in $^1O_2$ production was recorded in the optimized nanoparticle-modified systems. These observations are closely linked to the LSPR. In summary, these findings provide a robust framework for the design of nanostructured systems with improved therapeutic efficiency and resistance to photodegradation.

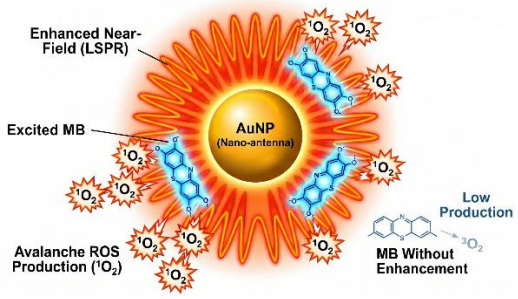


Fig. 1. Mechanism of plasmonic enhancement of reactive oxygen species (ROS) generation.
Illustration of LSPR-mediated near-field photoexcitation of methylene blue (MB) [Generated by Gemini 3.1 Pro based on user prompt]

Novel Electrocatalysts for CO_2 Conversion Based on Functionalised Carbon Materials

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Keywords: transition metals, electrocatalysis, CO_2 reduction reaction, oxygen evolution reaction

Electrochemical reduction of carbon dioxide (CO_2) represents a promising approach for converting CO_2 into valuable fuels and chemicals, such as carbon monoxide (CO), formic acid (HCOOH), and methanol (CH_3OH). This process can be conducted under ambient temperature and pressure conditions, powered by renewable energy sources. Compared to carbon capture and storage technologies, CO_2 reduction reaction (CO_2RR) potentially offers lower energy consumption, making it a more sustainable solution. Within a CO_2 electrolyser, CO_2RR occurs at the cathode, while the oxygen evolution reaction (OER) takes place at the anode. Both CO_2RR and OER are kinetically slow, multi-step electrochemical reactions, with CO_2RR competing against the hydrogen evolution reaction. To achieve desired chemical transformations with high selectivity and energy efficiency, suitable electrocatalysts are essential.

Noble metals like gold, iridium, and ruthenium have shown high catalytic activity for CO_2RR and OER. However, their high cost, limited availability, and low stability pose significant barriers to industrial applications. Consequently, recent research has shifted towards using cheaper, more abundant transition metals such as iron, nickel, and cobalt, along with minimising metal content in electrocatalytic materials [1–3]. Carbon materials serve as promising supports for metallic active sites, enabling high catalytic activity with minimal metal loading. The use of carbon cloth or carbon paper facilitates the synthesis of free-standing electrodes, eliminating the need for polymeric binders.

This work presents low-content transition metal electrocatalysts composed of highly dispersed active species on activated carbon supports. The electrocatalysts were prepared through electrodeposition from low-concentration baths of iron, nickel, or cobalt ions, using thermally oxidised carbon cloth or carbon paper as the support. The morphology and chemical composition of the materials were analysed, employing methods such as scanning electron microscopy, transmission electron microscopy, X-ray diffraction, X-ray photoelectron spectroscopy. The electrocatalytic activities of the resulting electrodes towards OER and CO_2RR were investigated, demonstrating the potential of these materials as inexpensive and efficient electrocatalysts for clean energy technologies.

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Influence of Vacuum Conditions on the Tribological Properties of Polymeric Materials

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Keywords: tribology, vacuum, polymers, solid greases, space industry

This work focuses on theoretical aspects of vacuum tribology and outlines the planned research activities for a first-year doctoral project. It introduces the concept of tribology and its importance in modern science. In addition, the main differences between classical and space tribology are discussed. Phenomena occurring in vacuum, such as outgassing or cold welding, and their influence on tribological properties [1], [2] are also analyzed.

Differences in friction coefficients for materials operating in vacuum and under normal atmospheric conditions are presented; for example, the dynamic coefficient of friction in metal-to-metal contacts can increase by three times in low vacuum [3]. The effect of the absence of oxygen on the surface layer structure and its impact on tribological properties [2] is also investigated.

Following the theoretical part, the research plan for the coming year is outlined. The method planned to investigate changes in the friction coefficient is the Pin-on-Disc. A dedicated test stand was constructed to operate under low vacuum conditions (1–100 Pa) with the applied load ranges from 10 N to 120 N. The study aims to develop friction coefficient characteristics of different material pairs in future work – for example 1.4404 steel with PEEK–MoS_2–WC self-lubricating composite. This combination is advantageous due to high PEEK thermal stability (compared to other polymers) and superior lubricating properties of MoS_2–WC mixture in vacuum environments [2].

In conclusion, this work provides a theoretical background for space tribology and proposes a scientific approach to investigate the influence of low vacuum on tribological properties of selected material pairs used in the space industry.

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From Algal Waste to Functional Biochar

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Keywords: algae biomass, thermochemical conversion, biochar, environmental protection, sustainable agriculture

Sustainable agriculture increasingly requires innovative strategies to improve soil quality while simultaneously addressing the growing challenge of biomass waste management. Biochar has gained significant attention as a soil amendment due to its ability to enhance soil structure, nutrient retention, and overall plant performance [1]. It is most commonly produced by pyrolysis (350–700°C) from lignocellulosic, agricultural and forest residues. However, alternative biomass sources, such as marine and freshwater macroalgae, remain largely unexplored despite their high availability and distinct physicochemical composition. At the same time, algal biomass often constitutes a problematic waste stream, particularly in coastal and aquatic environments. Moreover, excessive algal blooms are frequently associated with anthropogenic activities, including nutrient runoff from agriculture and wastewater discharge, leading to eutrophication of aquatic ecosystems and further emphasizing the need for effective valorization strategies.

In response to these challenges, this work aims to present the assumptions and methodological framework for exploring the potential of waste algae biomass as a raw material for the production of value-added materials, in particular biochar, with applications in agriculture and environmental protection, through the proposed research. Biochar will be produced from marine and freshwater macroalgae using two complementary thermochemical processes: torrefaction (200–300°C) and pyrolysis (300–600°C). Process parameters will be systematically optimized using a Design of Experiments (DoE) approach to achieve high yields and desirable material properties. To ensure a comprehensive understanding of the materials, both the raw algal biomass and the resulting biochars will undergo detailed physicochemical, structural, and functional characterization. The planned analyses include C (carbon) H (hydrogen) N (nitrogen) elemental analysis; multielemental composition anlyses using Inductively Coupled Plasma-Optical Emission Spectroscopy (ICP-OES) and X-ray fluorescence (XRF); Fourier Transform Infrared Spectroscopy (FTIR) for the detection of functional groups on the biochar surface; thermogravimetric analysis (TGA); Brunauer–Emmett–Teller analysis (BET), as well as pH, and electrical conductivity determination. These analyses will enable direct comparison of materials obtained via torrefaction and pyrolysis and support the identification of key factors influencing their performance. The obtained biochars will be evaluated in two main application pathways. First, their sorption properties will be investigated in the context of environmental remediation, including the removal of metal ions from aqueous solutions and the remediation of soils contaminated with heavy metals. To enhance sorption efficiency, selected materials will be further modified through chemical activation (e.g., KOH) as well as $MgCl_2$ or low-temperature plasma treatment. Second, biochars will be assessed as soil amendments, with a focus on their impact on soil physicochemical properties, including porosity, bulk density, pH, organic matter content, nutrient availability, cation exchange capacity, etc. These effects will be further examined in pot experiments, where the influence of algal biochar on plant growth will be evaluated through selected morphological, physiological, and biochemical parameters. Overall, this research aims to develop sustainable and multifunctional solutions for the valorization of algal biomass, contributing to soil improvement, pollution mitigation, and the advancement of circular economy principles.

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From Design to Desire: Investigating the Person – Environment Fit in Futuristic Cities

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Keywords: futuristic cities, urban experience, social habitus, human-centered urbanism

Rapid urban population growth has accelerated the development of highly engineered urban environments aimed at optimising environmental performance and quality of life. Such approaches, however, often rest on the assumption that people will naturally adapt to these designed conditions. Despite advances in human-centred urbanism, contemporary practice continues to privilege technological and environmental efficiency over social sustainability.

This tension is particularly evident in future-oriented urban developments. Projects such as Songdo and Masdar have struggled to attract residents and sustain social vitality despite their technological sophistication [1], while similar dynamics in China have resulted in so-called "ghost cities" [2]. These cases suggest that urban challenges may arise not only from economic or political conditions, but from a misalignment between urban form and human expectations.

Recent research increasingly points to the role of urban form in shaping social experience and well-being [3]. Urban environments operate as systems of spatial signals through density, layout, and materiality that individuals interpret in relation to their everyday practices [4]. This perspective shifts attention toward person–environment fit, suggesting that urban success depends on the alignment between spatial design and users' cognitive, behavioural, and emotional expectations.

This doctoral research examines how individuals perceive and evaluate different urban forms, and under what conditions they accept, adapt to, or reject them. It uses qualitative research methods to identify socially learned expectations that inform how individuals interpret urban environments, particularly in the design phase. The aim of this research is to elicit expectations, identify norms and develop taxonomy of urban experience. Achieving this understanding will reframe the success of future cities as a question of compatibility: cities do not fail when they cannot be built, but when they do not fit the people they are designed for. Ultimately, this PhD work aims to generate a research framework to study urban experiences in the design phase of new urban development plans and models.

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Mechanism of Metal Powder Formation in High-Power Ultrasonic Atomization with In-situ High-Speed Diagnostics

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Keywords: powder morphology, melt atomization, droplet breakup, ultrasonic wave

Ultrasonic atomization of metals is a promising method for producing high-quality powders for additive manufacturing and surface engineering applications [1], [2]. Despite growing interest in this approach, the fundamental mechanisms of droplet formation under high-frequency sonotrode vibrations remain insufficiently understood, particularly regarding process limits and material-dependent behavior [2], [3]. The aim of this study was the experimental identification and analysis of metal powder formation mechanisms in a high-power ultrasonic field.

The conducted research demonstrated that ultrasonic atomization enables the production of powders characterized by high sphericity (>0.95), narrow particle size distribution, uniform morphology, and preserved chemical stability [1], [4]. For stainless steel 316L, the particle size distribution at 20 kHz was characterized by d_{10} = 44.45 µm, d_{50} = 81.47 µm, and d_{90} = 146.8 µm, whereas at 40 kHz these values decreased to d_{10} = 40.28 µm, d_{50} = 61.41 µm, and d_{90} = 95.87 µm, indicating that higher excitation frequency promoted finer powder production and narrower size distributions. The integration of a continuous wire feeding system and plasma-based melting enabled stable process conditions and processing of a broader range of metallic materials, including newly designed alloys. High-speed imaging performed at acquisition rates exceeding 80,000 frames per second enabled direct observation of liquid metal behavior on the sonotrode surface and breakup dynamics at excitation frequencies of 20 and 40 kHz. This allowed identification of critical stages of droplet formation and experimental validation of theoretical models describing ultrasonic atomization.

The results revealed clear correlations between sonotrode vibration parameters and powder characteristics, showing that increased vibration amplitude and frequency promoted finer droplets and improved particle sphericity. Microstructural and chemical analyses confirmed the absence of significant contamination or compositional deviations, while highlighting the influence of process parameters on internal particle integrity. The study also identified physical and technological limitations related to vibration amplitude, melt stability, and droplet detachment mechanisms under high-frequency excitation.

These results deepen understanding of the complex phenomena occurring during ultrasonic atomization and provide an experimental basis for optimization and scaling of this technology, opening opportunities for controlled design of metal powders for additive manufacturing and thermal spraying [1], [2].

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Deep Learning for Automated Sleep Staging from Mouse EEG Recordings

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Keywords: EEG, sleep staging, deep learning, mouse models, neuroscience

Sleep analysis based on electroencephalographic (EEG) recordings is a fundamental tool in neuroscience, providing key insights into brain function, circadian regulation, and the mechanisms underlying neuropsychiatric disorders. In particular, the characterization of sleep architecture – encompassing wakefulness, non-REM, and REM stages, as well as associated neural oscillations such as delta and theta rhythms -offers a sensitive framework for detecting alterations induced by chronic stress and pharmacological interventions in animal models [1–4].

This study focuses on the development and application of advanced computational methods for automated EEG-based sleep staging in mice. Using high-quality, multi-channel recordings, including datasets such as the Mouse Sleep Staging Validation (MSSV) resource [5], we analyze electrophysiological signals to identify state transitions and quantify sleep structure with high temporal resolution. Special emphasis is placed on leveraging artificial intelligence techniques, particularly deep learning models, to improve the accuracy and scalability of sleep classification.

By prioritizing efficient and reproducible EEG analysis pipelines, this work aims to enable sustainable large-scale studies of sleep in animal models. The proposed approach supports the identification of subtle changes in sleep patterns associated with stress and therapeutic interventions, contributing to a deeper understanding of brain function and the development of more precise and environmentally responsible tools for neuroscience research.

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From Space to the ice edge: how SAR and AI automate the glacier calving front monitoring

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Keywords: AI, SAR, Glaciers, Calving Fronts, monitoring

Deep learning models and transformers are now widely utilized for detecting the position of glacier calving fronts. However, in most cases, these models operate on single-polarization SAR data, which makes them susceptible to difficulties in identifying ice mélange – a dense mixture of frozen seawater and icebergs that visually blends with the glacier terminus. Furthermore, SAR data has limited spatial resolution, ranging from ~20 × 20 m (IW) to ~40 × 40 m (EW), which effectively caps the accuracy of results obtained by both deep learning algorithms and human interpreters. While optical and multispectral data could offer alternatives, their use is restricted by the requirement for solar illumination, which is unavailable during the polar night.

To address the challenge of ice mélange identification, our research utilizes dual-polarization (HH and HV) satellite radar data (SAR). This approach enables a precise analysis of the backscatter structure, drastically increasing the contrast between the solid glacier and the surrounding mélange. To overcome current accuracy limitations and the existing 40–80 m margin of error, we are planning in-situ calibration research expeditions to Greenland and Svalbard. These campaigns are aimed at refining the capabilities of deep learning models through direct field measurements. Additionally, we are fine-tuning the Segment Anything Model 2 (SAM 2) using dual-polarization SAR and Harmonized Landsat and Sentinel-2 (HLS) optical data to investigate their influence on model performance.

Review of the Temperature Measurement Techniques Being Used in Pulsating Heat Pipes

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Keywords: pulsating heat pipe, temperature measurement, OFDR, distributed optical fiber sensor one

Temperature distribution, particularly its local variation under unsteady conditions, is a key parameter for understanding thermal flow processes in pulsating heat pipes (PHPs) a passive, high efficient, heat exchanger commonly use in satellites. However, commonly used experimental techniques exhibit significant limitations. Point-wise sensors, such as thermocouples, provide accurate measurements but only at discrete locations, making it impossible to reconstruct a continuous temperature profile and capture transient, localized thermal phenomena [1]. Infrared (IR) thermography, on the other hand, enables high-resolution mapping of the external surface temperature, but does not provide direct information about the internal fluid. Estimation of internal conditions requires solving an inverse heat conduction problem, which introduces additional assumptions and potential inaccuracies [2, 3].

To address these challenges, Distributed Optical Fiber Sensing (DOFS) based on Rayleigh scattering and Optical Frequency Domain Reflectometry (OFDR) is proposed as an advanced measurement technique. In this approach, a single optical fiber acts as a continuous sensor, enabling spatially distributed temperature measurements with millimeter-scale resolution along its entire length. Due to its small diameter, the fiber can be integrated directly within the PHP, allowing for direct measurement of the internal fluid temperature with minimal influence on the flow structure. Furthermore, the method supports high-frequency data acquisition, enabling detailed analysis of transient thermal and flow phenomena [4, 5].

As a result, DOFS provides spatially continuous and direct insight into the temperature field within PHPs, significantly improving the investigation of their complex and dynamic thermal behavior compared to conventional methods [6]. With over 1500 measurement points per meter, it enables detailed internal temperature mapping, improving efficiency evaluation and providing more reliable data for numerical analysis.

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Controlling Bifunctionalisation Pathways in Metalloporphyrin-Catalyzed C(sp^3)–H Chlorination of Arylalkanes

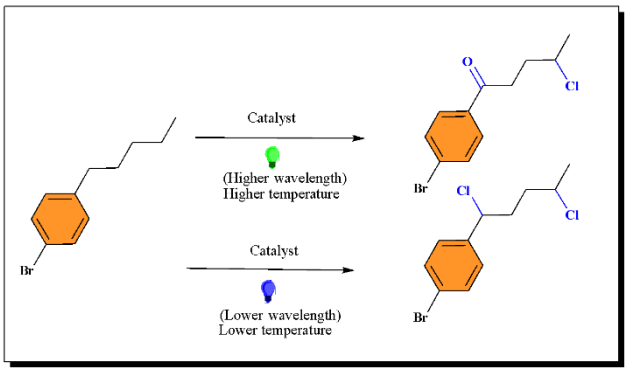
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Keywords: Metalloporphyrin catalysis, C(sp3)-H activation, bifunctionalisation, halogenation

Metalloporphyrins are widely studied as functional molecular systems for promoting alkyl C–H functionalisations, motivated by their structural similarity to enzymatic cofactors and their ability to operate under relatively mild conditions [1]. These complexes have been widely reported to be effective mediators for C(sp^3)–H halogenation [2, 3]. While selective bifunctionalisation of the alkyl chain in arylalkane systems is feasible, achieving high selectivity is hindered by parallel activation pathways that yield closely competing products. Following the principle of the previous work on the synthesis of 1,4-dicarbonyl compounds, in which it was hypothesised that water plays a significant role in selectivity [4], this project extends the chlorination protocol.

The role of metalloporphyrin complexes in this transformation under photocatalytic conditions is examined through a comparative, control-based approach. Various *meso*-tetraphenyl-porphyrin complexes incorporating different central metal ions (Mn, Cr, Ti, Co) were evaluated under analogous conditions, alongside reactions performed in the absence of metalloporphyrins. Initial results suggest that product formation is influenced by both the reaction environment and the presence of metalloporphyrin complexes, where slight variations in the reaction parameters alter the functional groups at the carbon centres (Scheme 1). At the same time, control experiments indicate that background reactivity can contribute significantly under certain conditions, highlighting the need for careful interpretation of catalytic effects. Current efforts focus on quantitatively optimising each conditions separately and identifying optimal conditions. GC-MS monitors the reactions and the products are quantitatively analysed by NMR spectroscopy.



Scheme 1. Representation of the slight variation in conditions leading to the two major products in chlorination

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Mobile Biomimetic Robots for Extreme Space Environments Exploration

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Keywords: biomimetics, robotics, space exploration, automation, mobile robots

Space exploration presents a set of different challenges for rovers and space probes, the need to survive rocket launch, while minimizing mass and size, travel through vacuum utilizing hibernation mode in harsh thermal conditions and high radiation levels to existing conditions on target celestial bodies, if the mission profile focuses on exploring those. Low gravity, dusty environment on Mars, vacuum, fine regolith, high temperature fluctuations and two week long night on the Moon, high atmospheric pressure on Venus, loose rocks and gravel in microgravity on various comets and asteroids are just a few examples of drastically different parameters that future space missions will have to cope with.

Life on Earth presents a high degree of adaptability to different environmental conditions. Evolutionary process, in the timespan of over 4,5 billion years has led to a development of various traits essential to survival. From single-cell extremophiles focused on survival strategies to animals with different mobility solutions, organisms have colonized almost every environment on the planet. Each organism is essentially a separate prototype testing different adjustments to these traits against environmental conditions on Earth. Engineers can therefore utilize various biological inspirations to help conquer the final frontier for life, which is space environment. However, since no organism known currently to humans has evolved to survive and thrive in outer space, biological solutions cannot be simply copied, they need to be fully understood to propose their uses and to compare their performance to more traditional space systems.

This presentation will focus on showcasing a few short examples of biological inspirations for space exploration. Focus will be on potential of improving climbing on steep walls, like Mars canyons by taking inspiration from a hoof of an Alpine ibex [1], leaping in low gravity conditions to improve navigation in Moon craters, similar to grasshoppers or frogs [2] and improved digging mechanisms in microgravity asteroid mining, by modeling a wood wasp ovipositor [3].

Taking inspiration from biological organisms presents a way to improve parameters of space missions, open new environments and make space exploration cheaper and more sustainable. These advancements made possible by better understanding of biology, novel materials and increase in computational power allowing for better modeling and machine learning showcase potential for a new age in space industry.

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The Role of Corruption in Human Trafficking

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Keywords: human trafficking, corruption, international criminal law

Human trafficking, often described as a form of modern-day slavery, is a crime consisting in the recruitment, transport, transfer, harbouring or receipt of a person by such means as threat or use of force or other forms of coercion, abduction, fraud or deception for the purpose of exploitation. It is a violation of fundamental human rights and freedoms such as the right to life, liberty and personal security. Over the last years trafficking in persons has reached an ever-increasing scale. There is one factor which immensely contributes to the high level of human trafficking – corruption. A crime which is both an underlying root cause and a facilitating tool to carry out and sustain this illicit trade. Unfortunately, there has been very little exploration into the linkages of human trafficking and corruption. Corruption is a "lifeline of the traffickers" [1] as it reduces their risks of operating. At the same time, it indicates a very serious weakness in the rule of law in particular countries, to which an independent judiciary and an impartial legal system are vital elements [2]. Among those countries perceived to be the most corrupt, human trafficking efforts are extremely low, and governments are not making significant efforts to improve[3]. And since corruption likely exists in all countries to some capacity, the identification and investigation of human trafficking cases can be severely undermined by corrupted officials. Corruption not only plays a significant role in the modus operandi of traffickers but also instils fear in victims of trafficking, fuels distrust toward law enforcement officials, and constrains reporting to authorities who are the very entities implicated in the trafficking enterprise[4]. Corruption related to human trafficking can involve the public sector, private sector or both of them at the same time. It can be observed at the stage of searching for potential victims, transporting them and during their exploitation. Corruption of public officials related to human trafficking may be also divided depending on the role of such officials. He or she may be: a trafficker, facilitator of human trafficking or facilitator of impunity[5].

Corruption facilitates trafficking at all stages, destabilises democracy, weakens the rule of law and hinders economic development. Simultaneously trafficking itself contributes to the high level of corruption as to function trafficking strongly relies on bribed police officers, prosecutors, border guards, immigration officers, judges, politicians and other public officials, but also persons from private sector like entrepreneurs and travel or employment agencies. If it weren't for corruption, human trafficking would not have expanded so rapidly. It allows the trafficking process to remain protected from prosecution, assists the victim's movements within a country and across borders and furthermore helps perpetrators hide the illegal profits gained from human trafficking. Therefore, any type of anti-trafficking policies and strategies should fully incorporate a focus on preventing corruption. Actions such as: raising awareness of the links between corruption and human trafficking; improvement of detection, investigative and enforcement capacity to combat corruption in human trafficking; targeting resources to high risk areas; punishing corrupted officials; improving transparency and quality of employment practices (especially those concerning public officials); improving data collection and research regarding human trafficking and corruption but also establishing transparent public, private and institutional procurement and acquisition systems and encouraging the private sector to develop effective internal controls should be taken to prevent both corruption and human trafficking.

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SAR-Based Approaches for Automated Detection of Grounding Line Dynamics

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Keywords: SAR, glaciers, grounding line, Antarctica

Research on ice sheets has intensified in recent decades due to the need to better understand and predict the effects of ongoing climate change. In particular, accurate estimation of ice mass loss is essential for determining the amount of freshwater flux to the oceans, which affects sea-level rise, ocean salinity, and ocean circulation. These processes play a key role in the Earth's climate system and in global heat distribution.

One of the most important indicators of ice shelf dynamics is the grounding line, defined as the boundary where a glacier detaches from the bed and begins to float. Among the available detection methods, Differential Interferometry of Synthetic Aperture Radar (DInSAR), particularly when combined with the double-difference technique, provides one of the most accurate results [1, 2]. Recent advances in machine learning algorithms further enhance this method by automating grounding line recognition [3], significantly reducing the processing time required for results generation.

The long-term time series allows for the identification of the minimum and maximum extents of grounding line migration, which in turn enables the estimation of grounding zone width, representing the true magnitude of glacier movement. The conducted research reveals seasonal variability, with grounding line retreat typically occurring toward the end of the year and re-advance around mid-year. Additionally, a high degree of similarity in migration extremes was observed across different zones of the same glacier, consistent both in timing and magnitude.

The development of a new automated method based on SAR amplitude images may further advance research on glacier change. Unlike traditional approaches requiring multiple SAR acquisitions, this method does not rely on correlations between consecutive observations and enables grounding line positioning at the exact time of a single satellite pass. This approach has the potential to improve both the accuracy and temporal resolution of grounding line migration analysis.

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Influence of Roadside Vegetation Structure on Traffic Noise Attenuation Along the S5 Expressway

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Keywords: traffic noise, roadside vegetation, green infrastructure, noise attenuation, expressways

Road traffic noise is one of the most significant environmental stressors affecting areas located near major transportation corridors. Increasing traffic intensity and the expansion of high-capacity road networks require effective and environmentally sustainable noise mitigation strategies. Roadside vegetation belts are increasingly recognized as a viable complement to conventional acoustic barriers, as they can influence sound propagation through absorption, scattering, and diffraction of sound waves while also providing ecological and landscape benefits.

The aim of this study was to assess the influence of roadside vegetation structure on traffic noise attenuation along a section of the S5 expressway in Poland. Acoustic measurements were conducted in July 2025 at four research sites located along a 1.5 km segment between the Poznań Wschód interchange and the Kleszczewo exit in the direction of Gniezno. The sites differed in the structure and composition of roadside vegetation, enabling a comparative assessment of its impact on noise propagation.

Site S1 represented a control area without woody vegetation and consisted only of herbaceous plants. Site S2 included a 10 m wide belt of shrubs (*Cornus alba* "Elegantissima") located approximately 15 m from the road edge and reaching heights of 2.5–3 m. Site S3 consisted of a mixed shrub belt composed of *Physocarpus opulifolius* and *Sambucus nigra*, planted alternately in several rows and forming a vegetation strip approximately 10.8 m wide. Site S4 represented a multilayer vegetation structure consisting of three rows of shrubs (*Spiraea* × *vanhouttei*) followed by regularly spaced trees (*Acer pseudoplatanus*) reaching approximately 7.5 m in height.

The study is based on field measurements conducted using two class 1 sound level meters installed at fixed distances from the road edge (9 m and approximately 25–26.5 m). A-weighted equivalent sound pressure levels (LAeq) were recorded with a time resolution of 1 s over repeated 5-minute intervals, resulting in 1-hour measurement series per site. Frequency-dependent sound levels were additionally analysed in one-third octave bands (20 Hz–20 kHz). The analysis focuses on the relationship between vegetation structure, including belt width, species composition and vertical stratification, and the level of traffic noise attenuation between measurement points. The comparison of the results obtained from the measurement sites allows for the evaluation of how different vegetation configurations influence sound propagation. As the study is ongoing, it is hypothesized that vegetation belts characterized by greater structural complexity and multilayered composition provide more effective acoustic attenuation than areas without woody vegetation.

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Balancing Simplicity and Control in Catalyst Deposition for Carbon Nanostructure Growth on Carbon Cloth

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Keywords: carbon nanostructures, carbon cloth, electrocatalysis, Zn-air batteries

The performance of zinc-air batteries is strongly limited by the sluggish kinetics of the oxygen reduction (ORR) and oxygen evolution (OER) reactions. Carbon nanostructures (CNSs), directly grown on conductive substrates, offer a promising route toward binder-free and efficient air electrodes [1]. The aim of this study is to develop a simple and scalable catalyst deposition strategy that enables controlled growth of CNSs with tunable properties, directly influencing their ORR and OER electrocatalytic performance.

A dip-coating method based on aqueous solutions of transition metal salts was employed for catalyst deposition, followed by catalytic chemical vapor deposition (CCVD) for in situ CNS synthesis. The obtained materials were comprehensively characterized and tested as ORR/OER electrocatalysts. The results demonstrate that the dip-coating approach ensures uniform catalyst distribution and reproducible CNS growth, leading to enhanced electrocatalytic activity toward both reactions compared to untreated carbon cloth, confirming its suitability for application as air electrodes in Zn-air batteries.

To further improve control over catalyst properties, additional deposition techniques were explored, including pulsed laser ablation (PLA) and additive nanoparticle patterning. While PLA resulted in the formation of ultrasmall catalyst nanoparticles that did not promote effective CNS growth under the applied CCVD conditions, additive nanoparticle patterning enabled tunable catalyst loading [2]. Preliminary gravimetric studies revealed a clear correlation between precursor concentration and carbon yield after CCVD, with higher concentrations leading to increased mass gain.

These findings identify dip-coating as a robust, low-cost, and scalable method for the fabrication of high-performance carbon-based electrodes, while advanced deposition strategies provide additional control over catalyst characteristics. This combined approach opens new opportunities for the rational design of efficient air electrodes for Zn-air batteries.

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Uncertainty and Limitations in the Modelling of Semi-Transparent BIPV Systems in Building Energy Simulations: A Review

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Keywords: BIPV glazing, building energy simulation, model uncertainty, model quality, BIM–BEM interoperability

Semi-transparent Building-Integrated Photovoltaic (BIPV) glazing systems represent complex façade solutions in which optical, thermal, and electrical phenomena are inherently coupled. Their implementation in Building Energy Simulation (BES) requires the simultaneous representation of solar radiation transmission, heat transfer, and electricity generation, which increases model complexity and sensitivity to input data [1].

Previous studies indicate that simulation results are highly dependent on input parameters and modelling assumptions, particularly those related to glazing properties such as visible transmittance and solar heat gain coefficient [1, 2]. Moreover, it has been demonstrated that different simulation tools may produce significantly divergent results even when applied to the same building model, highlighting the systemic nature of uncertainty in BES analyses [3].

The aim of this study is to review current approaches to modelling BIPV glazing and to identify the main sources of uncertainty affecting simulation outcomes. The literature analysis indicates that input data quality, modelling simplifications, and model preparation procedures play a critical role in determining the reliability of results.

Recent research also emphasizes that interoperability between BIM environments and simulation tools remains a significant challenge. The transformation of models between these environments often requires geometric simplifications and data reduction, which may lead to information loss and inaccuracies in simulation outputs [4].

The presented review provides a foundation for further research aimed at improving the reliability of BIPV simulations and supporting the development of methodological guidelines for model preparation and validation.

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Towards High-Temperature Spin Qubits in Telecom Quantum Dots

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Keywords: single quantum dots, telecom spectral range, spin dephasing, Hanle effect

Quantum photonic technologies rely on the generation of non-classical states of light, such as single photons and polarization-entangled photon pairs, to enable secure communication and distributed quantum computing. Semiconductor quantum dots (QDs) have emerged as one of the leading solid-state platforms for realizing such sources, offering scalability and compatibility with existing fiber-optic infrastructure. In particular, QDs emitting in the telecommunications C-band are highly suited for long-range quantum networks; however, their performance is often limited by carrier spin dephasing induced by nuclear spin fluctuations, which degrades spin coherence and, consequently, the fidelity of entangled states [1, 2].

We investigate single InAs(P)/InP QDs grown by molecular beam epitaxy using a ripening technique that enables low areal density (~2×10^9 cm^{-2}) and improved in-plane symmetry. A distributed Bragg reflector enhances photon extraction efficiency up to 13.3% [3], while photon correlation measurements confirm high-purity single-photon emission with $g^{(2)}(0) < 0.01$ [4] and the generation of polarization-entangled photon pairs via the biexciton–exciton cascade [5].

Spin depolarization is studied using Hanle measurements under quasi-resonant circularly polarized excitation in a magnetic field up to 0.3 T (Voigt geometry) [6]. The sign of the degree of circular polarization (DOCP) depends on the excitation energy, indicating distinct carrier relaxation pathways [7]. With increasing temperature (20–60 K), the zero-field DOCP decreases from ~70% to ~20%. At the same time, the electron spin dephasing time T_2^* is reduced from 1.7 ns to 0.6 ns. A finite DOCP at high magnetic fields suggests partial nuclear spin polarization.

Despite the large nuclear spin of indium, the observed coherence times are comparable to GaAs-based systems [8], likely due to the larger QD volume. These results demonstrate that telecom QDs can sustain spin coherence at elevated temperatures, highlighting their potential for spin-based quantum technologies.

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Sustainable Regeneration of Graphite from Spent Lithium-Ion Batteries Using Organic Acids

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Keywords: lithium-ion batteries, graphite recycling, organic acids, green chemistry, electrode regeneration

The increasing demand for lithium-ion batteries (LIBs) creates an urgent need for sustainable recycling strategies. While current processes mainly target cathode metals, graphite anodes remain insufficiently recovered despite their critical raw material status and high share in battery mass [1, 2]. Previous studies have shown that organic acid leaching, including citric acid treatment, can enable graphite recovery and reuse from spent LIBs [3]. In this study, spent graphite from 18650 LIB cells was regenerated using thermal pretreatment followed by acid leaching with organic acids as greener alternatives to conventional mineral acids.

Spent cells were discharged and dismantled, and graphite was separated from copper anode foils. After thermal treatment at 500 °C, the material was leached with acetic, ascorbic, citric, and malic acid, while sulfuric acid was used as a conventional reference. The regenerated graphites were characterised by XRD, SEM, BET, FTIR, and ICP-OES. Their electrochemical behaviour was evaluated in lithium half-cells using galvanostatic cycling, cyclic voltammetry, and rate capability tests.

The results showed that thermal pretreatment removed crystalline SEI-related phases, whereas acid leaching influenced surface cleanliness, impurity removal, and electrochemical stability. Citric acid produced the highest structural ordering, while ascorbic acid gave the most stable cycling behaviour among the tested organic acids. Sulfuric acid removed the highest amount of impurities but caused strong electrochemical instability, showing that aggressive leaching does not necessarily improve electrode performance.

Overall, citric and ascorbic acids were identified as the most promising leaching agents, offering a favourable balance between structural preservation, purification, and electrochemical functionality. These results demonstrate the potential of organic acid-based graphite regeneration as a sustainable route for LIB recycling and circular battery material use.

Acknowledgments

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Novel Chemical Probes for Imaging Protease and Kinase Activity

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Keywords: chemical activity-based probes, DNA-binding dyes, tyrosine kinase inhibitors, enzyme imaging, mass cytometry

Proteases and kinases participate in numerous biological processes, making them critical components of living cells [1]. Consequently, a wide range of methods for detecting and analyzing enzymatic activity has been developed. One such approach involves chemical probes – compounds consisting of a recognition element and a reporter tag. These tools can be divided into substrate-based probes, in which signal amplification occurs after enzyme-mediated transformation, and inhibitor-based probes, which bind directly to enzymes, enabling their spatial localization (Fig. 1) [2]. However, existing chemical probe strategies face several limitations. Insufficient signal amplification reduces sensitivity, while poor cellular retention and low membrane permeability restrict their use in live-cell applications. This underscores the need for novel chemical tools to study enzyme activity.

Here, we report on the design, synthesis, and biological evaluation of novel probes for investigating the activity of medically relevant caspases (–1, –6, –8) and kinases (BTK, HER family). These include caspase-selective substrate-based probes with DNA-binding dyes. In these probes, both fluorescence intensity and intracellular retention increase significantly upon binding to DNA, enabling real-time visualization of protease activity in living cells.

We also report on kinase-targeting probes obtained by transforming small-molecule tyrosine kinase inhibitor (TKI) moieties into metal-tagged probes to make them compatible with mass cytometry by time-of-flight (CyTOF) – a powerful modality for visualizing individual enzyme activity in different cell populations. Together, these tools expand the available strategies for studying enzyme activity.

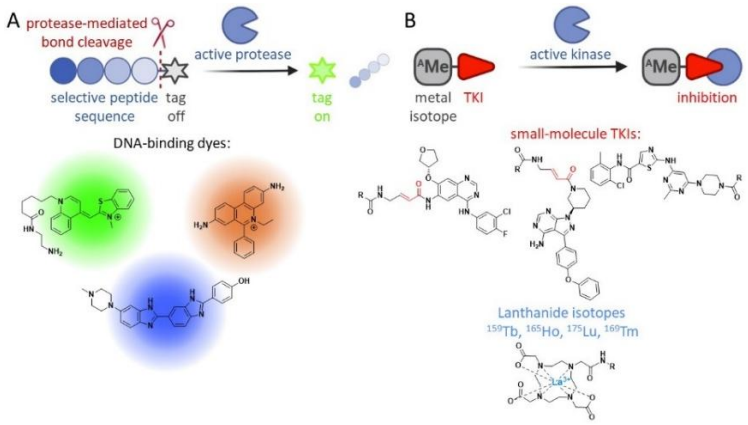


Fig. 1. (A) DNA-binding substrate-based probes for proteases, (B) mass cytometry inhibitor-based probes for kinases

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How to Obtain Luminescent Carbon?

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Keywords: carbon dots, graphene quantum dots, photoluminescence, hydrothermal process

Carbon dots (CDs), especially graphene quantum dots (GQDs), are a rapidly developing class of carbon nanomaterials due to their strong photoluminescence, chemical stability, and low toxicity. These properties make them highly promising for applications in optoelectronics, bioimaging, and sensing. The choice of synthesis method and precursor materials is crucial, as it determines the size, surface chemistry, and optical properties of the dots [1]. In this work, GQDs were synthesized using a one-pot green hydrothermal method with graphene oxide as a precursor and hydrogen peroxide as an oxidizing agent [2]. The resulting dispersions were purified by dialysis and then converted into solid form by lyophilization. The optical properties of the GQDs were investigated using UV-Vis absorption spectroscopy and photoluminescence spectroscopy, while the surface chemical composition and distribution of functional groups were analyzed by X-ray photoelectron spectroscopy (XPS). Additionally, Fourier transform infrared (FTIR) spectroscopy was used to identify surface functional groups and confirm the chemical structure of the GQDs, while high resolution transmission electron microscopy (HRTEM) was employed to determine their size and morphology. GQDs exhibit strong green photoluminescence with excitation-dependent emission behavior. XPS analysis confirms a high content of oxygen-containing functional groups, thereby highlighting their influence on the photoluminescence of the dots. Additionally, HRTEM analysis confirms that the size distribution of GQDs showed that the majority of GQDs are ~2 nm in diameter, followed by ~1 nm, with a minor fraction around ~3nm. To further evaluate their optical properties and identify potential applications, the quantum yield (QY) will also need to be determined. Overall, this work explains the influence of the synthesis method on size and surface chemistry of GQDs, thereby determining their optical properties and, consequently, their potential applications.

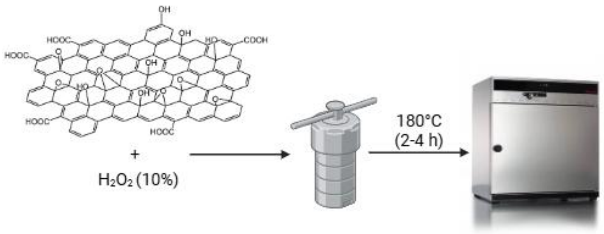


Fig. 1. Schematic illustration of synthesis of GQDs

Acknowledgments

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Human-Centred and Nature-Sensitive Street Design: Addressing the Gap Between Sustainability Policy and Practice

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Keywords: street design, human-centred urban planning, climate adaptation, sustainable urban mobility, nature-sensitive design

Urban streets are among the most extensive and enduring elements of the built environment, shaping mobility, environmental performance, and the quality of public life. Despite wide-ranging policy commitments to climate adaptation, biodiversity, and liveable cities, street design in practice remains largely shaped by technocratic road-engineering standards and the spatial legacy of motonormativity. The growing prominence of "smart city" narratives – centred on digitalisation, vehicle electrification, and autonomous transport – does not resolve this tension: technological optimisation applied to inherited spatial structures leaves car-dependent configurations largely intact and inadequately addresses challenges of climate resilience, biodiversity loss, and social inclusivity.

This research addresses the gap between sustainability agendas and the processes through which streets are actually planned and delivered. It asks how human-centred and nature-sensitive design practices – rather than technological upgrading alone – can effectively respond to contemporary challenges of climate change, urban heat, biodiversity decline, and the social costs of car-dependent environments. Building on urban design theory, environmental psychology, and project management frameworks inspired by iterative development models, the study develops a conceptual model embedding behavioural research, stakeholder engagement, staged testing, and post-implementation evaluation as structural components of the design process. The goal is to understand how interdisciplinary integration can reorganise street transformation into a coherent, adaptive, and participatory process – one capable of translating policy ambitions into consistent spatial outcomes rather than isolated successful exceptions.

The empirical foundation draws on an ongoing case overview based on analysis of project documentation, municipal strategies, and technical reports – sources that constitute the primary available literature on these recent transformations. The overview encompasses street redesign projects in Utrecht, Vienna, Copenhagen, Amsterdam, Paris, and Barcelona, examining principles of street reallocation, public participation, sustainable mobility, and climate adaptation. Two projects provide detailed evidence of integrated technological and ecological innovation: the Nieuwe Mark project in Breda (Netherlands), where nature-inclusive quay walls with species-specific mortar and dimensioned joints actively support mosses, ferns, insects, and birds, developed through iterative laboratory testing and demanding on-site coordination, as well as circadian-responsive lighting systems reducing energy consumption by over 60%; and the Rue Garibaldi redevelopment in Lyon (France), where self-sustaining mixed-species planting addresses urban heat, stormwater management, and wildlife habitat simultaneously. In each case, outcomes depended on sustained interdisciplinary coordination and adaptive project processes rather than on individual technologies. The case overview is further extended by an in-depth case study of street transformations in Amsterdam, Paris, and Barcelona, currently underway.

Preliminary analysis of the case overview material suggests that closing the gap between sustainability policy and street-level practice depends less on the availability of technical solutions and more on governance capacity to align spatial, ecological, social, and organisational dimensions from the earliest project stages. Human-centred and nature-sensitive approaches demonstrate clear potential – but remain exceptional rather than standard, pointing to the need for reframing street transformation as a continuous, evidence-informed, and interdisciplinary process. The research aims to contribute to understanding the procedural and governance pathways through which such reframing can enable systemic change in street design, rather than isolated transformation.

Exit of a “Corporate Prisoner” from a Spółka z o.o. (Polish Limited Liability Company)

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Keywords: corporate prisoner, limited liability company, shareholder, exit of a shareholder

The subject of the doctoral thesis currently being prepared by the author, which he intends to briefly outline in his presentation, is the issue of the “corporate prisoner” in a spółka z o.o. (Polish limited liability company). The term “corporate prisoner” refers to a shareholder who would like to end their membership of the company, but for legal or practical reasons this proves impossible, meaning that such a shareholder must continue their membership of the company against their will. The reasons behind a decision to exit a company can be very diverse. They may relate to the shareholder’s personal reasons (e.g., retirement), to relations with other shareholders (e.g., conflict with other shareholders) or to the shareholder’s relations with the company (e.g., a prolonged failure to distribute profits, infringement of the shareholder’s rights). In practice especially minority shareholders can find themselves in the position of a “corporate prisoner”, who not only have no realistic chances of finishing their membership of a company, but also cannot influence the decisions taken at the general meetings of shareholders, and their rights as shareholders might be infringed. In such cases such a position is particularly disadvantageous for such a shareholder.

In the forthcoming doctoral thesis the author analyses the various mechanisms, that can be used by the shareholder as a “corporate prisoner” in order to exit the company, including the possibility for a shareholder to dispose of all their shares, as well as the dissolution of the company by court resulting from a shareholder’s action. Additionally, consideration is given to the possibility of introducing a specific procedure for a shareholder’s exit from the company while maintaining the company’s existence.

The significance of this research lies in the fact that the limited liability company is the most commonly chosen form of commercial company for conducting business activities within a company in Poland.

Process Parameter Screening for PBF-LB/M of Zn–Mg Biodegradable Alloys: Preliminary Study

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Keywords: PBF-LB/M, bioresorbable implants, lightweight metals, magnesium, zinc

Additive manufacturing based on the PBF-LB/M process offers new opportunities for processing biodegradable metals such as magnesium and zinc, which are considered promising materials for temporary biomedical implant applications [1]. Thanks to their controlled degradation in physiological conditions, these materials allow the creation of products that are fully resorbed, eliminating the need for surgical removal and minimizing the risk of complications. However, processing Mg and Zn using the PBF-LB/M method presents several challenges, including intense evaporation, difficulties in controlling chemical composition and the formation of defects such as porosity, lack of fusion and microcracks [2].

This preliminary study focuses on process parameter screening for PBF-LB/M processing of Zn–3Mg powder blends. The investigated processing window included laser power (40–46 W), scanning velocity (200–230 mm/s), exposure time (200 µs), point distance (40–46 µm), and hatch spacing (65 µm), using a chess scanning strategy. A Zn–3Mg powder mixture was prepared using a V-type mixer to ensure homogeneity. Cubic samples were manufactured using a Realizer SLM 50 system. The specimens were mounted in resin and analysed using confocal microscopy. Porosity was quantified via image-based analysis.

The highest relative density of 97.94% was achieved for a volumetric energy density (VED) of 101.10 J/mm^3. This indicates that within the investigated parameter window, higher energy input improves melt pool stability and densification. However, residual porosity is still present, suggesting incomplete defect elimination likely related to vaporization effects and melt pool instability typical for Zn-based alloys. From a functional perspective, this density level is expected to improve mechanical performance by increasing load-bearing area and reducing stress concentration sites. At the same time, residual porosity may influence corrosion behaviour by promoting localized degradation, which could be relevant for controlled resorption applications. Further work will focus on predictive modelling to optimize process parameters and establish relationships between processing conditions, microstructure, mechanical properties, and corrosion behaviour.

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Small Changes, Big Consequences: Can Local Chemistry Predict Protonation?

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Keywords: pKa, protonation, computational chemistry, machine learning, electron density

Why does a molecule give away a proton or keep it? It sounds like a simple decision, but it plays a key role in many chemical processes, from drug action to catalysis. In principle, acidity follows from the underlying electronic structure, yet predicting pKa in practice is far from straightforward [1], especially for larger molecules with several possible protonation sites.

This research takes a closer look at a simple idea: maybe not all parts of a molecule matter equally. The working hypothesis is that the key factors controlling protonation are mostly encoded in the immediate environment of the protonation site, while more distant regions may have a smaller, though not negligible, influence. Whether this "local view" is sufficient remains an open question.

To explore this, the project combines computational chemistry with machine learning. Interpretable models are used to highlight physically meaningful trends, while neural networks help capture more subtle patterns in molecular structure and electron density. Looking at both side by side makes it possible to see when a local description works and when it starts to break down.

The goal is not only to predict pKa values, but also to better understand what controls them. This can be useful for designing molecules in areas such as drug discovery, catalysis, and materials science, where even small shifts in protonation can make a noticeable difference.

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